



ACKNOWLEDGMENTS

This Training is supported by a research grant funded by the Research, Development, and Innovation Authority (RDIA), Kingdom of Saudi Arabia, with grant number 13382-psu-2023- PSNU-R-3-1-EI.

Prof. Ahmad Taher Azar

Full Professor, Leader of Automated Systems & Computing Lab (ASCL)

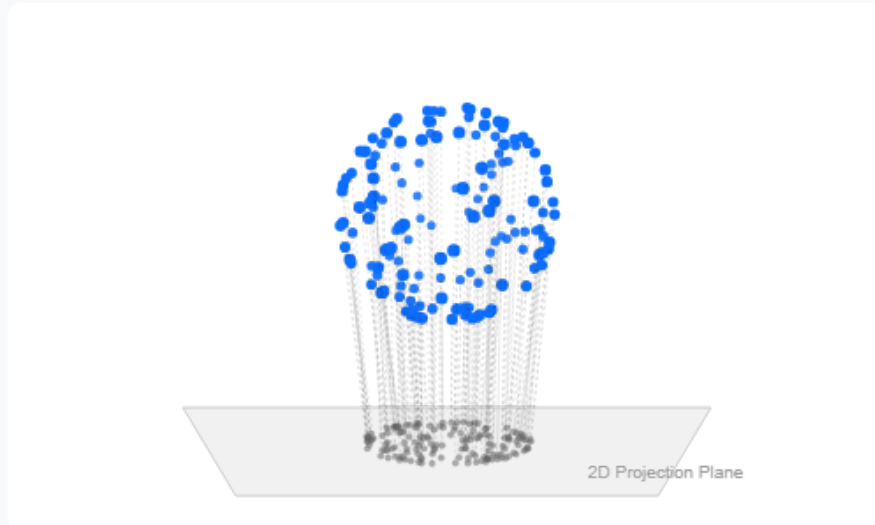
Principal Component Analysis (PCA)

1. The Intuition: The Shadow of Data

PCA is about simplification. Imagine a 3D object casting a 2D shadow.

If you rotate the object correctly, the shadow preserves most of the object's shape. If you rotate it poorly (e.g., looking at a plate edge-on), you lose all the information.

Visualization: 3D Sphere Point Cloud to 2D Plane



Blue Points: Original 3D Data (Sphere)

Gray Points: Projected 2D Data (Shadow)

Dashed lines show the mapping from 3D to 2D

- **Goal:** Find the "best angle" (Principal Component) to view the data so that the points are as spread out (variance) as possible.
- **Result:** Reduce 100 confusing variables down to 2 or 3 key themes.

Visualizing Dimensionality Reduction ($n \rightarrow k$)

Imagine a single data point (like one customer) described by 5 features (Age, Income, Debt, Score, Location). We want to compress this to just 2 meaningful numbers.

Original Input Vector ($n=5$)

x_1
x_2
x_3
x_4
x_5

→
PCA Transformation

Compressed Vector ($k=2$)

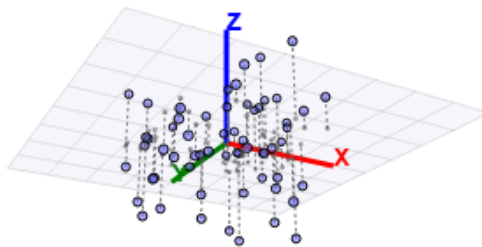
PC_1
PC_2

We turn a "tall" vector of 5 numbers into a "short" vector of 2 numbers.
Each new number (PC) is a weighted mix of **all** original numbers.

Interactive 3D Demo

Visualizing data in 3 Dimensions (X, Y, Z). The animation rotates to show how 3D data might look projected onto a 2D plane (shadow).

Red: X-axis, Green: Y-axis, Blue: Z-axis.



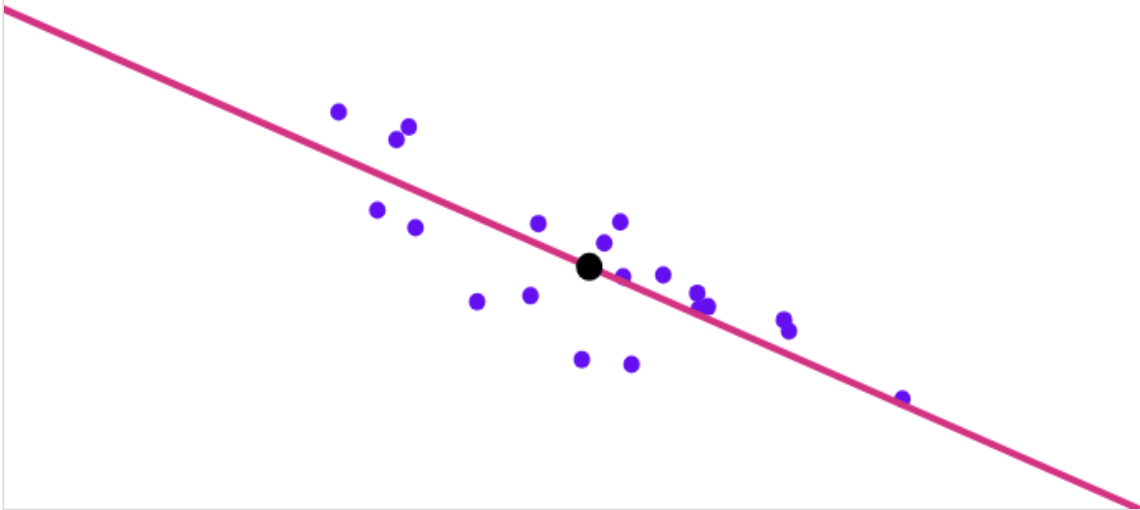
2. Visualization (2D): Finding the Main Axis

Click to add points. The Purple line shows the 1st Principal Component (direction of max variance).

Basic Math: The 1st Principal Component is the unit vector $\mathbf{w}$ that maximizes the variance of the projected data points:

$$\mathbf{w}^* = \arg \max_{\|\mathbf{w}\|=1} \mathbf{w}^T \mathbf{S} \mathbf{w}$$

where $\mathbf{S}$ is the covariance matrix. This vector $\mathbf{w}^*$ turns out to be the eigenvector of $\mathbf{S}$ with the largest eigenvalue.



Interactive: Why Variance?

Rotate the blue line (direction $\mathbf{w}$) to see how the "spread" (variance) of the projected points changes. The **Current Variance** you see below is calculated as $\mathbf{w}^T \mathbf{S} \mathbf{w}$.

Math Connection: PCA doesn't guess angles like we are doing here. It solves the calculus problem to find where the slope of variance is zero:

$$\frac{\partial}{\partial \mathbf{w}} (\mathbf{w}^T \mathbf{S} \mathbf{w}) = 0 \implies \mathbf{S} \mathbf{w} = \lambda \mathbf{w}$$

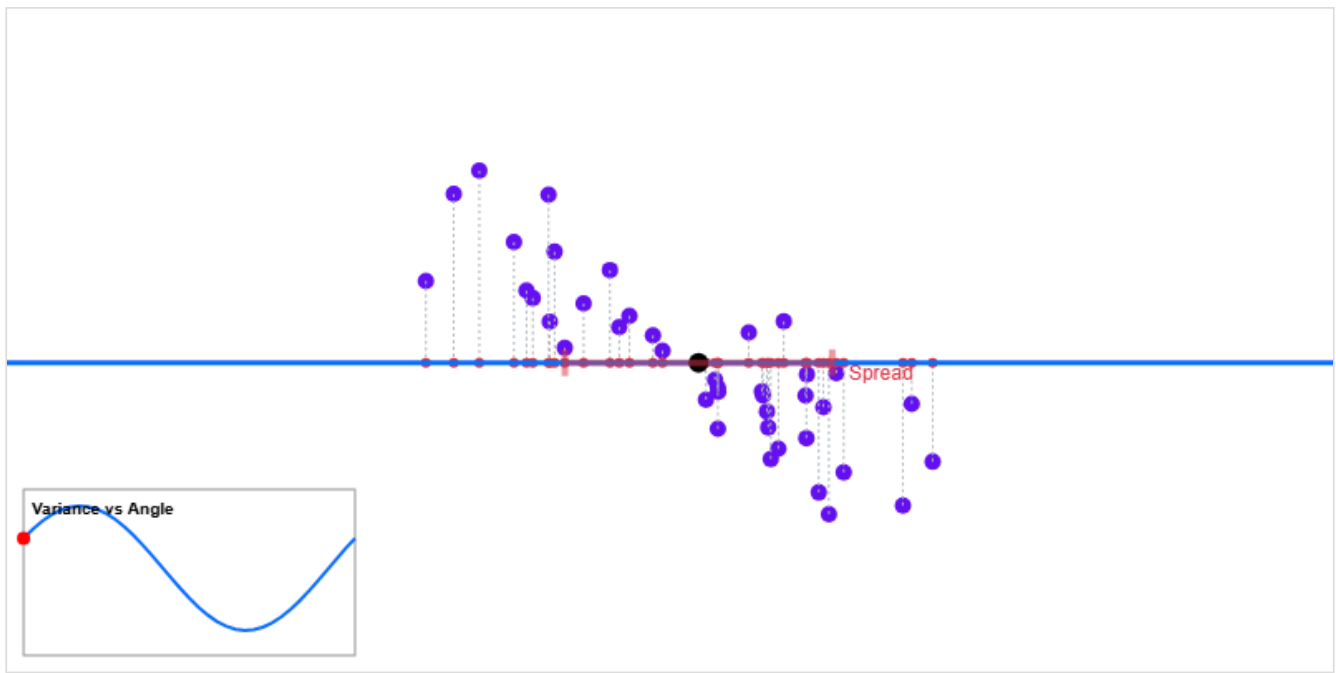
This is why the direction of maximum variance is exactly the **Eigenvector** with the largest Eigenvalue.

Rotation Angle:



Current Variance: 6503

Max Possible Variance: 8594



3. The Math: Eigenvalues & Eigenvectors

We want to find a direction u that maximizes the variance of the projected data.

The Optimization Problem

$$\max_{u: \|u\|=1} u^T S u$$

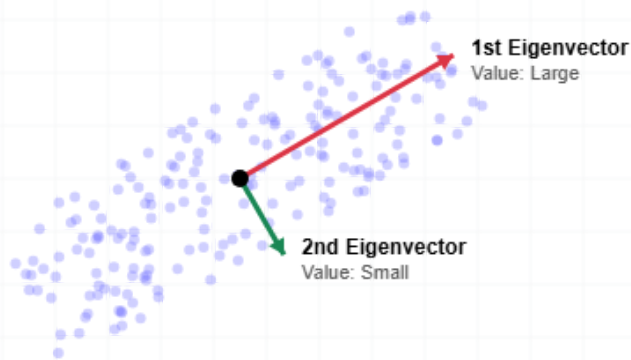
Where S is the Covariance Matrix of the data. We are looking for the vector u that captures the most spread.

The Solution

Think of **Eigenvectors** as "Directions" and **Eigenvalues** as "Importance".

- **1st Principal Component (Eigenvector 1):** The direction where data varies the most (like the length of a baguette). It has the largest Eigenvalue.
- **2nd Principal Component (Eigenvector 2):** The next most important direction, which must be perpendicular to the first (like the width of a baguette). It has the second largest Eigenvalue.

Simple Rule: We sort Eigenvectors by their Eigenvalues (high to low) and pick the top k to be our new axes.



4. Step-by-Step PCA Walkthrough

Let's trace the PCA process with a tiny dataset to see exactly how the numbers work. We have **3 samples** and **3 features** (F1, F2, F3), and we want to reduce it to **2 dimensions**.

Step 1: The Dataset

Sample	Feat 1	Feat 2	Feat 3
A	4	1	1
B	2	3	5
C	6	8	0
Mean	4	4	2

Step 2: Center the Data

Subtract the mean from each value so the new mean is 0.

Sample	F1 - 4	F2 - 4	F3 - 2
A	0	-3	-1
B	-2	-1	3
C	2	4	-2

Step 3: Covariance Matrix

We calculate how features vary together using the formula: $\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$. Since we have $n = 3$ samples, we divide by $3 - 1 = 2$.

Calculation Examples:

- **Variance of F1** (Top-left): $\frac{(0)^2 + (-2)^2 + (2)^2}{2} = \frac{0+4+4}{2} = 4$
- **Covariance of F1 & F2** (Top-middle): $\frac{(0)(-3) + (-2)(-1) + (2)(4)}{2} = \frac{0+2+8}{2} = 5$

$$S = \begin{bmatrix} 4 & 5 & -5 \\ 5 & 13 & -4 \\ -5 & -4 & 7 \end{bmatrix}$$

Step 4: Eigenvalues & Eigenvectors

We need to find the vectors v such that $Sv = \lambda v$. We start by solving the characteristic equation $\det(S - \lambda I) = 0$.

1. Find Eigenvalues (λ):

Set the determinant to zero:

$$\det \left(\begin{bmatrix} 4 - \lambda & 5 & -5 \\ 5 & 13 - \lambda & -4 \\ -5 & -4 & 7 - \lambda \end{bmatrix} \right) = 0$$

Solving this cubic polynomial gives roots: $\lambda_1 \approx 20, \lambda_2 \approx 3, \lambda_3 \approx 1$.

2. Check Variance Conservation:

Total Variance (Trace of S) = $4 + 13 + 7 = 24$.

Sum of Eigenvalues = $20 + 3 + 1 = 24$. (Matches!)

3. Calculate Explained Variance Ratio:

How much information does each PC hold?

- **PC1:** $\frac{20}{24} \approx 83.3\%$
- **PC2:** $\frac{3}{24} \approx 12.5\%$

The corresponding Eigenvectors (directions) are:

- **PC1 (Major Axis):** $v_1 \approx [0.3, 0.8, -0.5]$
- **PC2 (Minor Axis):** $v_2 \approx [-0.6, 0.5, 0.6]$
- PC3 (Discarded): $v_3 \approx [0.7, 0.3, 0.6]$

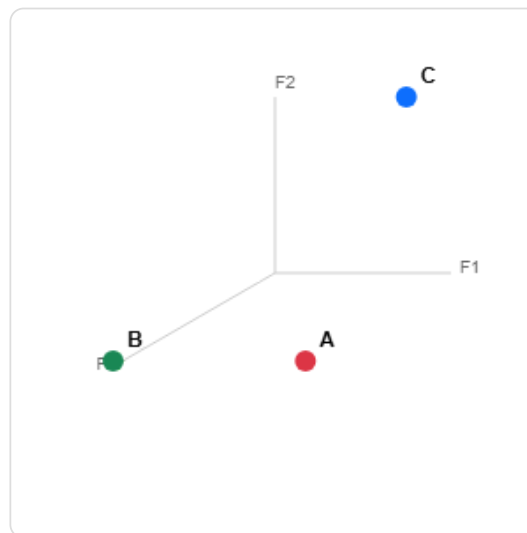
Step 5: Projection (3D to 2D)

We take our centered data matrix (3×3) and multiply it by our chosen feature vectors (3×2).

$$\text{New Data} = \begin{bmatrix} 0 & -3 & -1 \\ -2 & -1 & 3 \\ 2 & 4 & -2 \end{bmatrix} \times \begin{bmatrix} 0.3 & -0.6 \\ 0.8 & 0.5 \\ -0.5 & 0.6 \end{bmatrix} = \begin{bmatrix} -1.9 & -2.1 \\ -2.9 & 2.5 \\ 4.8 & -0.4 \end{bmatrix}$$

Result: We started with 3 features (columns) and now have 2 features. Sample A is now at coordinates $(-1.9, -2.1)$ in the new PC space.

Original 3D Space (F1, F2, F3)



New 2D Space (PC1, PC2)



5. Python Implementation

Using **PCA** from scikit-learn to reduce Iris data from 4D to 2D.

```
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
```



```

def pca_demo_plot():
    """
    PCA demo on Iris dataset.
    """

    # 1. Load Data
    data = load_iris()
    X = data.data
    y = data.target

    # 2. Scale Features (CRITICAL for PCA!)
    # PCA is sensitive to scale, so we must standardize first.
    scaler = StandardScaler()
    X_s = scaler.fit_transform(X)

    # 3. Apply PCA
    pca = PCA(n_components=2)
    Z = pca.fit_transform(X_s)

    # 4. Check results
    print("Explained variance ratio:", pca.explained_variance_ratio_)
    # e.g., [0.72, 0.23] means 1st PC explains 72% of variance, 2nd explains 23%.

    print("\nPrincipal Components (Feature Weights):")
    print(pca.components_)
    # PC1 = w1*SepalLen + w2*SepalWid + w3*PetalLen + w4*PetalWid
    # These weights tell us which original features matter most for each axis.
    # e.g. If PetalLen has a high weight in PC1, then PC1 is mostly measuring
    Petal Length.

    # 5. Plot
    plt.scatter(Z[:, 0], Z[:, 1], c=y, cmap='viridis')
    plt.xlabel("First Principal Component")
    plt.ylabel("Second Principal Component")
    plt.title("Iris Dataset Projected to 2D")
    plt.show()
pca_demo_plot()

```